FeatherLM

FEATHERLM · FIREWORKS DEPLOY HACKATHON

Copy that reads like a person wrote it, keeps every fact, at a tenth of the cost.

An open model (GLM-5.3 Flash on Fireworks) inside a harness that checks its own work. $0.81 per 1,000 rewrites against $9.11 for Claude Opus 5.5.

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THE PROBLEM

Ask a frontier model to "make it sound human" and it edits your facts.

DRAFT (DEV-092)

Tagline: Break Language Barriers & Unlock Nordic Markets 🌍

CLAUDE OPUS 5.5

Tagline: Localize Your SaaS for the Baltic and Nordic Markets

Gate G2: "You dropped these names: barriers, break, hey, language, unlock."

FEATHERLM (O3C)

Tagline: Break Language Barriers & Reach Nordic Markets

First comment: Hi, I'm Kristjan, founder of Linguo Labs. There are 14 of us, working across 4 countries. …

Gates G1–G4: pass, 0 failed.

Across 247 test items, Claude Opus 5.5 keeps every fact in **71%** (176/247). The other 29% ship with something changed.

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HOW IT WORKS

A cheap open model, held to account by code.

1 · Draft

GLM-5.3 Flash on Fireworks rewrites the AI draft in plain prose.

→

2 · Check

G1 nothing invented · G2 nothing lost (numbers, names, quotes) · G3 no AI tells · G4 not gutted (0.7–1.3× length).

→

3 · Targeted fix

Failed check → exact instruction ("You dropped these names… put them back"). Up to 3 rounds.

→

4 · Ship

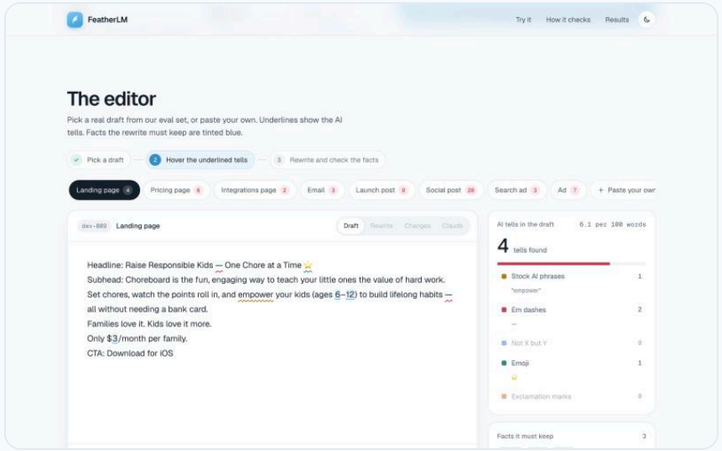
Only a rewrite that passes all four counts as fact-safe.

Same checker scores FeatherLM and every baseline. Median rewrite time 3.4 s (Claude: 3.7 s).

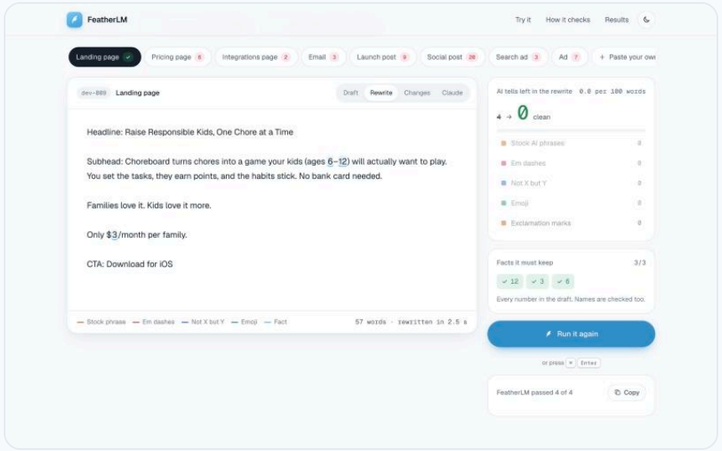
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THE PRODUCT · HOW YOU USE IT

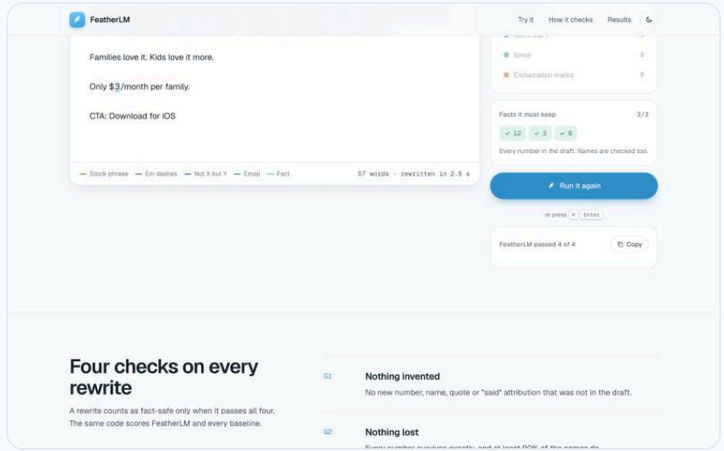
Paste a draft. See the tells. Get a rewrite that proves it kept the facts.



1 Pick or paste a draft. AI tells are underlined (4 here); facts to keep are tinted blue.



2 Rewrite. Tells 4 → 0, facts 3/3 kept (12, 3, 6).



3 "FeatherLM passed 4 of 4." Copy it out.

Blind reader test live at `featherlm-ab.vercel.app/round2/`

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BENCHMARK · FACT-SAFE REWRITE RATE (STRICT)

FeatherLM keeps the facts 97% of the time. Claude: 71%.

SYSTEM · 247 ITEMS	FACT-SAFE		$ / 1K
FeatherLM (O3c, GLM-5.3 Flash + harness)	**97%** 239/247		**$0.81**
Claude Opus 5.5 + our harness	94% 231/247		$15.85
Claude Opus 5.5 (strong prompt)	71% 176/247		$9.11
Claude Opus 5.5 + humanizer prompt	60% 147/247		$50.22
GPT-6.1 Terra	40% 98/247		$0.85
GLM-5.3 Flash, no harness	27% 66/247		$0.10

Out of sample, **80 real READMEs**: FeatherLM **97.5%** (78/80) vs Claude **76%** (61/80). Δ vs Claude on 247: +26 pp [+20, +31], p<0.001 (paired bootstrap).

HOW WE MEASURED

One scorer for every system, plus checks on the scorer.

GATES (FACT-SAFE = ALL 4)

- G1 nothing invented
- G2 nothing lost
- G3 no AI tells
- G4 length 0.7–1.3×

Strict and lenient (forgives regex artefacts). 95% CIs, paired bootstrap Δ vs Claude.

SETS

- 130 eval (100 dev + 30 trick)
- 117 held-out real human pairs
- 80 real READMEs, out of sample
- Control: copy the draft = 32%

CHECKS ON THE CHECKS

- 2 LLM judges, both orders (Kimi K3, Qwen 3.8 Max)
- Spot the human vs real originals
- Blind human A/B (209 votes)
- Anti-gaming audit: copying, dropped sentences

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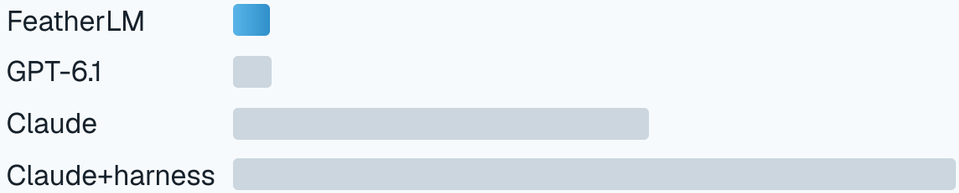
BUSINESS · WHAT IT COSTS AT VOLUME

Same job, about 11× cheaper than Claude. 20× cheaper than Claude doing it our way.

REWRITES / MONTH	FEATHERLM	CLAUDE OPUS	CLAUDE + HARNESS	GPT-6.1
10k	$8	$91	$159	$9
100k	$81	$911	$1,585	$85
1M	$806	$9,111	$15,850	$853
10M	$8,061	$91,107	$158,502	$8,534

GPT-6.1 costs the same as us but is fact-safe only 40% of the time.

MONTHLY COST AT 1M REWRITES



25,506

broken-fact rewrites avoided per 100k vs Claude Opus 5.5 (strict: 28.7% fail vs 3.2%).

$/1k from results.json cost_per_1k_usd: O3c $0.806, Claude strong $9.111, F1H Claude+harness $15.850, GPT-6.1 Terra $0.853. Avoided = (71/247 − 8/247) × 100,000. Lenient gate: 13,360 per 100k (15.4% vs 2.0% fail).

STYLE · THE HONEST PART

On style, readers leaned to Claude. Not significant, and we know why.

BLIND HUMAN A/B · FROZEN 15:15

FeatherLM preferred	64
Claude preferred	89
Both fine	56

209 votes · 45 readers · sign test p = 0.052. Leaned to Claude, not significant.

LLM JUDGES SPLIT

qwen3p8-max, style: 44 ours / 27 Claude / 37 tie (p = 0.057)
kimi-k3, style: 33 ours / 46 Claude / 51 tie (p = 0.177)

Judges agree with human majorities on only 4/10 (kimi) and 3/10 (qwen) items.

Readers' note: ours over-uses dramatic fragments and punchy one-line closers. Claude writes calm complete sentences.

ABLATIONS · WHAT MOVED THE NUMBER

The loop did the work. Training helped the base model, not the facts.

STEP	TECHNIQUE	STRICT	$/1K
O1	GLM-5.3 Flash, plain	27%	$0.10
O2	+ shared system prompt	73%	$0.31
O3	+ harness: gates → targeted fix, ≤3 rounds	93%	$0.28
O3c	**+ prompt + style round (entry)**	**97%**	**$0.81**
O3k	+ claim checker, best-of-3	99%	$3.18
E2	Ensemble GLM + DeepSeek	100%	$3.41
R3	Same harness on DeepSeek	95%	n/a
SFT	Fireworks Training API, reverse-distilled licence-clean Common Pile data	26.7 → 32.8 → 37.7%	–

BUILT TODAY

What exists, what we tried, what's next.

SHIPPED

- Harness: 4 code gates + ≤3 targeted fix rounds
- Editor: tells, fact tints, diff, Claude compare
- Blind A/B page on Vercel
- Benchmark: 247 items + 80 real READMEs

FINE-TUNES

- GLM SFT clean: 38%
- GLM SFT modern: 33%
- Qwen3 4B DPO + harness: 8% (100 items)

Training alone didn't keep facts. The checks did.

NEXT

- Fix the style tell readers named (punchy fragments)
- Fine-tune with the harness in the loop
- API for teams shipping copy at volume

HONEST LIMITS

What this does not show.

- Readers did not prefer our style: 64 vs 89 (p = 0.052), leaned to Claude.
- Part of the strict margin is the gates' regex penalising harmless edits. Fair number: lenient, 98% vs 85%.
- 10% of harness (O3) outputs are near-copies of the draft (Claude: 1%).
- Gates check facts by matching; they don't prove meaning is unchanged.
- Fine-tuned models were not evaluated inside the harness on Fireworks (serving needed a paid deployment).
- Costs are per-token list prices from our runs, not a bill.

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We didn't make an open model smarter than Claude. We made it accountable, and made accountability about 20× cheaper.

$15.85 (Claude + our harness) ÷ $0.81 (FeatherLM) per 1,000 rewrites = 19.7×.

Sources: hackathon/report/results.json (systems[].cost_per_1k_usd; blocks.all and blocks.real pass/pass_lenient) · hackathon/report/HEADLINE.md (247-item table, fine-tunes, judges) · hackathon/report/human_votes.json (frozen 15:15) · hackathon/report/EXAMPLES.md + evals/runs/hack_O3c_glm_prompt_harness_style.jsonl (dev-092) · Screens: hackathon/demo/app/index.html, featherlm-ab.vercel.app/round2/

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APPENDIX A1 · ALL 31 ROWS, 247 ITEMS (EVAL 130 + HELD-OUT 117)

SYSTEM	STRICT [95% CI]	Δ STRICT VS CLAUDE STRONG	LENIENT	Δ LENIENT	$/1K
E2_ensemble_glm_deepseek	100% [98, 100]	+28 pp [+23, +34] *	100% [98, 100]	+15 pp [+11, +20] *	$3.41 (247/327 rows)
E1_ensemble	99% [97, 100]	+28 pp [+22, +34] *	100% [98, 100]	+15 pp [+11, +20] *	$15.12 (247/327 rows)
O3k_glm_calm_claimcheck_bo3	99% [97, 100]	+28 pp [+22, +34] *	100% [99, 100]	+15 pp [+11, +20] *	$3.18
O3l_glm_style_claimcheck_bo3	98% [96, 100]	+27 pp [+21, +33] *	100% [99, 100]	+15 pp [+11, +20] *	$3.06
O3j_glm_bestof3_judged	98% [95, 99]	+27 pp [+21, +32] *	99% [96, 100]	+14 pp [+10, +19] *	$2.61
O3m_glm_plainclaim	97% [94, 99]	+26 pp [+20, +32] *	99% [97, 100]	+15 pp [+10, +19] *	$0.91
O3t_glm_ali_taste	97% [94, 99]	+26 pp [+21, +32] *	100% [98, 100]	+15 pp [+11, +20] *	$0.89
O3c_glm_prompt_harness_style	**97% [94, 99]**	**+26 pp [+20, +31] ***	**98% [95, 99]**	**+13 pp [+9, +18] ***	**$0.81**
O3e_glm_harness_cliche	96% [93, 98]	+25 pp [+19, +31] *	99% [96, 100]	+14 pp [+10, +19] *	$0.81
O3f_glm_calm	96% [92, 98]	+24 pp [+19, +30] *	98% [96, 100]	+14 pp [+9, +18] *	$0.93
O3d_glm_harness_anticopy	95% [92, 97]	+24 pp [+18, +30] *	98% [95, 99]	+13 pp [+8, +18] *	$0.85
O3n_glm_frame	95% [92, 97]	+24 pp [+18, +30] *	96% [93, 98]	+12 pp [+8, +16] *	$0.95
R3_deepseek_harness	95% [91, 97]	+23 pp [+18, +29] *	99% [96, 100]	+14 pp [+10, +19] *	n/a
O3b_glm_prompt_harness	94% [91, 97]	+23 pp [+17, +29] *	97% [94, 99]	+12 pp [+7, +17] *	$0.54
R3t_deepseek_ali_taste	94% [91, 97]	+23 pp [+18, +29] *	98% [95, 99]	+13 pp [+9, +17] *	n/a
F1H_opus_harness	94% [90, 96]	+22 pp [+17, +28] *	97% [94, 99]	+13 pp [+8, +17] *	$15.85
O3_glm_harness	93% [89, 96]	+21 pp [+16, +27] *	97% [94, 99]	+12 pp [+7, +17] *	$0.28
O3g_glm_calm_retrieval_claimcheck	92% [88, 95]	+21 pp [+15, +27] *	97% [94, 99]	+13 pp [+8, +17] *	$1.42
O2_glm_prompt	73% [67, 79]	+2 pp [-4, +9], n.s. (p=0.535)	80% [74, 85]	-5 pp [-10, +0], n.s. (p=0.081)	$0.31
frontier-strong:claude-opus-5-5	71% [65, 77]	reference	85% [80, 89]	reference	$9.11
F1h_opus_humanizer	60% [53, 66]	-12 pp [-17, -6] *	77% [71, 82]	-8 pp [-13, -3], sig. (p=0.001)	$50.22
O2h_glm_humanizer	53% [46, 59]	-19 pp [-26, -11] *	63% [56, 69]	-22 pp [-28, -15] *	$1.15
frontier-live:claude-opus-5-5	53% [46, 59]	-19 pp [-24, -13] *	77% [71, 82]	-8 pp [-13, -3], sig. (p=0.003)	$7.17
F2h_gpt61_humanizer	51% [45, 57]	-20 pp [-27, -13] *	56% [50, 63]	-28 pp [-35, -22] *	$7.73
F2_gpt61_terra	40% [34, 46]	-32 pp [-39, -24] *	44% [38, 51]	-40 pp [-47, -34] *	$0.85
O4c_glm_sft_clean	38% [32, 44]	-34 pp [-41, -26] *	43% [36, 49]	-42 pp [-49, -35] *	n/a
O4m_glm_sft_modern	33% [27, 39]	-38 pp [-46, -31] *	41% [35, 47]	-44 pp [-51, -37] *	n/a
X0_copy_draft (no rewrite: control)	32% [26, 38]	-40 pp [-47, -32] *	32% [26, 38]	-53 pp [-60, -46] *	n/a
O1_glm_base	27% [21, 33]	-45 pp [-52, -37] *	30% [24, 36]	-55 pp [-62, -49] *	$0.10
frontier-persona-preexisting (130 items)	12% [7, 19]	-57 pp [-66, -48] *	16% [10, 24]	-66 pp [-74, -58] *	n/a
L2h_qwen3_4b_dpo_harness (100 items)	8% [4, 15]	-61 pp [-71, -50] *	9% [4, 16]	-73 pp [-82, -64] *	n/a

* significant (p=0.000, paired bootstrap). Missing rows count as fails. Source: hackathon/report/HEADLINE.md ← results.json.

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APPENDIX A2 · HELD-OUT REAL-HUMAN PAIRS AND 80 REAL READMES (STRICT)

SYSTEM	ALL 247	HELD-OUT 117	HELD-OUT CLEAN 96	REAL READMES 80
O3c_glm_prompt_harness_style	**96.8%** **239/247**	**98.3%** **115/117**	**99.0%** **95/96**	**97.5%** **78/80**
E2_ensemble_glm_deepseek	99.6% 246/247	99.1% 116/117	99.0% 95/96	98.8% 79/80
O3k_glm_calm_claimcheck_bo3	99.2% 245/247	99.1% 116/117	99.0% 95/96	98.8% 79/80
F1H_opus_harness	93.5% 231/247	94.9% 111/117	95.8% 92/96	97.5% 78/80
frontier-strong:claude-opus-5-5	71.3% 176/247	73.5% 86/117	71.9% 69/96	76.2% 61/80
frontier-live:claude-opus-5-5	52.6% 130/247	65.8% 77/117	64.6% 62/96	–
F1h_opus_humanizer	59.5% 147/247	65.8% 77/117	61.5% 59/96	–
F2h_gpt61_humanizer	51.0% 126/247	67.5% 79/117	65.6% 63/96	–
F2_gpt61_terra	39.7% 98/247	61.5% 72/117	59.4% 57/96	–
O4c_glm_sft_clean	37.7% 93/247	56.4% 66/117	62.5% 60/96	–
O4m_glm_sft_modern	32.8% 81/247	49.6% 58/117	51.0% 49/96	–
X0_copy_draft (no rewrite: control)	31.6% 78/247	56.4% 66/117	53.1% 51/96	–
O1_glm_base	26.7% 66/247	39.3% 46/117	33.3% 32/96	–

Held-out = real human originals never used in tuning; "clean" drops items flagged in heldout_excluded.json / train overlap. — = system not run on that block. Source: results.json blocks.

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APPENDIX A3 · LLM JUDGES AND SPOT-THE-HUMAN

PAIRWISE JUDGES, BOTH ORDERS · FEATHERLM (O3C) VS CLAUDE STRONG

JUDGE	OVERALL OURS/CLAUDE/TIE	STYLE OURS/CLAUDE/TIE
Qwen 3.8 Max	39/43/25 (p=0.741)	44/27/37 (p=0.057)
Kimi K3	35/52/43 (p=0.086)	33/46/51 (p=0.177)

Judge vs human majority (17 items): Kimi agrees 4 / disagrees 6; Qwen 3 / 7. Judges' style verdicts are not validated against readers.

SPOT THE HUMAN · HELD-OUT ORIGINALS VS REWRITE, BOTH ORDERS

SYSTEM	KIMI FOOLED / OR-SPLIT	QWEN FOOLED / OR-SPLIT
FeatherLM O3c	**0/117 · 27%**	**1/117 · 42%**
Claude strong	0/117 · 24%	0/117 · 34%
Claude + harness	0/117 · 21%	0/117 · 37%
GLM base	0/117 · 23%	0/117 · 27%
Draft copied (control)	0/117 · 14%	0/117 · 13%

Neither judge reliably picks any rewrite over the real human text.

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APPENDIX A4 · ANTI-GAMING AUDIT (HACKATHON/AUDIT/AUDIT.MD)

Is the harness gaming its own metric? Partly, and we say so.

SIGNAL	VERDICT	EVIDENCE
Gate fairness	harness wins	Gap to best frontier +21 pp; after forgiving regex artefacts still +11 pp (97% vs 85%)
Copying the draft	partly gaming	Harness (O3) near-copies 10% (Claude strong 1%); identical 0%; GLM base without loop 7%
Dropping sentences	harness wins vs Claude	Deleted-sentence rate 3.3% vs Claude strong −9.38 pp [−11.55, −7.45]; tie vs GPT-6.1
Artefacts	harness wins	Artefact share 1%; trailing numbers 0%; repeats 0%
Hold-out tells	mixed	In-band share 68.1%; vs Claude strong −4.23 pp (worse); human originals 87.5%
What the loop adds	disclosed	First-draft pass 23% → final 93% (O3); loops 0/1/2/3 = 57/105/61/27

"The fair number to quote is the lenient one" (AUDIT.md): lenient 98% vs Claude 85%. Do-nothing control (copy the draft) scores 32% strict.

APPENDIX A5 · TRAINING ON THE FIREWORKS TRAINING API AND LOCALLY

RUN	WHAT	RESULT (STRICT, 247)
Base	GLM-5.3 Flash, no harness (O1)	26.7%
SFT B (O4m)	Serverless LoRA, licence-clean reverse-distilled pairs; loss 1.144 → 0.699 (last-10 mean)	32.8%
SFT C (O4c)	Same recipe, gate-clean data; valid loss 1.116 → 0.752 (ppl 3.05 → 2.12)	37.7%
DPO → human refs (O7)	800 pairs, warm start from SFT B; serverless reference sampler hung at step 1	no model
Local DPO (L2, Qwen3 4B, MLX)	DPO toward real human writing on top of L1; loss saturates in ~20 steps	8% (100 items, with harness)
DPO-RLAIF (O6)	66 AI-labelled pairs + script prepared	not trained

SFT v1 (O4) trained but not evaluated (LoRA serving needed an on-demand deployment). Training lifts the open model, but not near the harness: the checks keep the facts.